

R et Rust : une histoire d'amour naissante

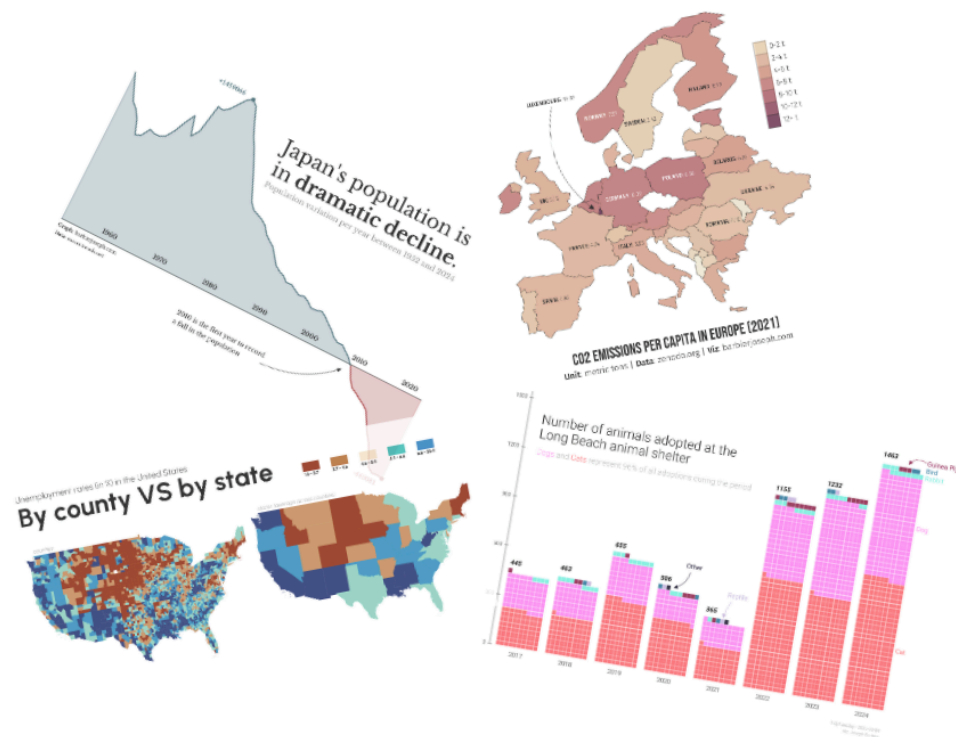
Joseph Barbier - *Rencontres R 2026*

A propos



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Consulting
Open source



Pourquoi venir parler de Rust ?

- *Pas pour vous dire que c'est mieux que R*
- *Pas pour vous dire que vous devez re-écrire vos projets en Rust*

Interest over time ⓘ

Worldwide · May 1, 2014 - May 24, 2026



Pourquoi venir parler de Rust ?

 **r/rust** • 1y ago
PsychologicalWall192

Thinking of Switching from C++ to Rust for Fun Projects – Is It Worth It?

Hey [r/rust](#)!

I've been using C++ for a while now, mainly for fun projects like fluid simulations, raytracers, and procedural map generation. While I love the performance and flexibility, I feel like I spend more time battling dependency issues and CMake than actually coding. It's been frustrating, and I'm starting to wonder if there's a better way.

Since I'm not tied to C++ for work and mostly code for fun, I'm considering making the switch to Rust. My main question: **Can Rust fill the niche that C++ does for me?** Specifically, I need a language that offers high-performance computing while being easier to set up and maintain.

For those of you who've used Rust for performance-critical applications like simulations, how has your experience been? And as someone coming from C++, how steep is the learning curve?

I'd love to hear your thoughts before diving in. Thanks in advance!

 157 

 73



 Share

 **marisalovesusall** • 1y ago

it's so much better you have no idea

yes and it's a no-brainer

 220 

 Reply

 Award

- **Open source** : Linux, Firefox, Deno, ...
- **Industrie** : Windows, AWS, Dropbox, Cloudflare, Discord, Datadog, ...

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▲ Why the developers who use Rust love it so much (stackoverflow.blog)
135 points by cyber1 on June 6, 2020 | hide | past | favorite | 165 comments

▲ nickm12 on June 6, 2020 | next [-]

I love this quote from a friend of mine who was learning Rust:

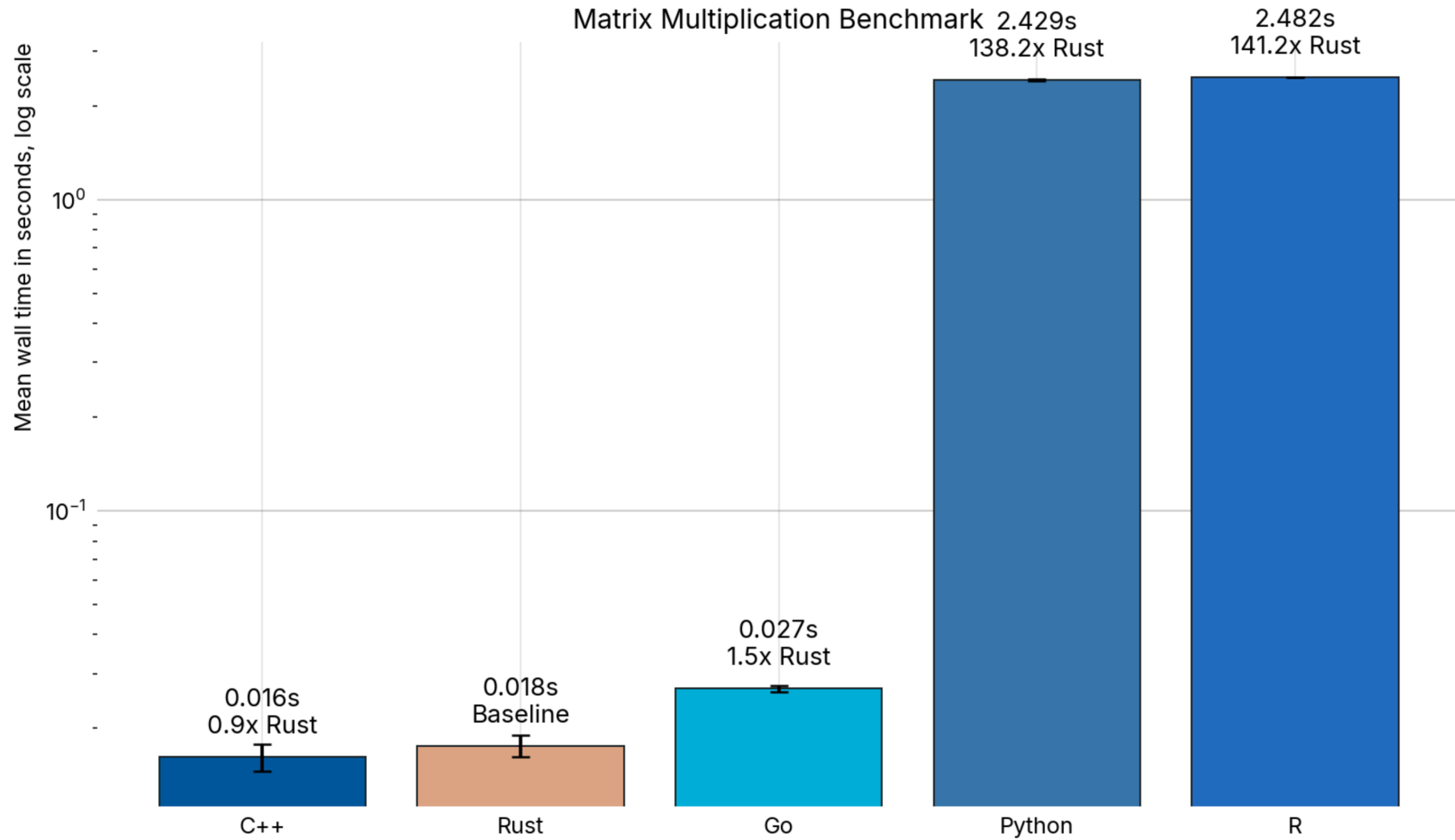
"It's hard but I love it. Dealing with the compiler felt like being the novice in an old kung fu movie who spends day after day being tortured by his new master (rustc) for no apparent reason until one day it clicks and he realizes that he knows kung fu."

Pourquoi venir parler de Rust ?

- 1. Comprendre les outils qu'on utilise
- 2. Donner envie d'essayer

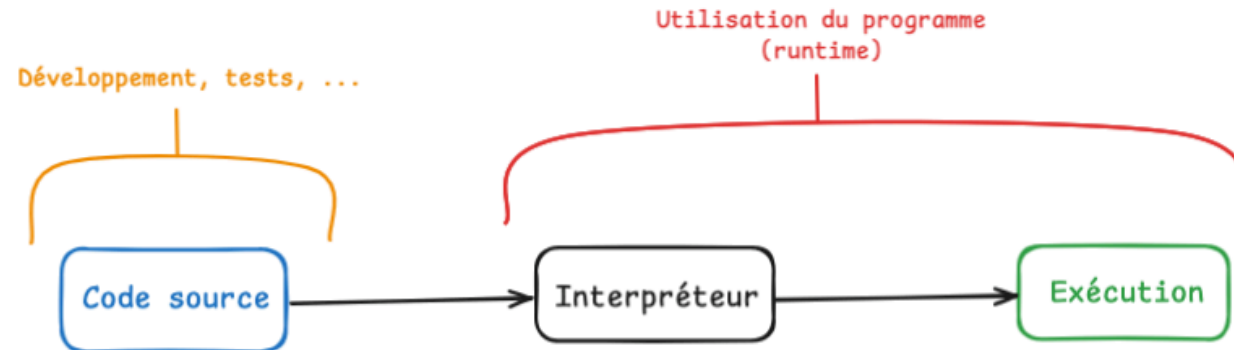
D'où ça vient ?

Performance (⚠️⚠️⚠️)

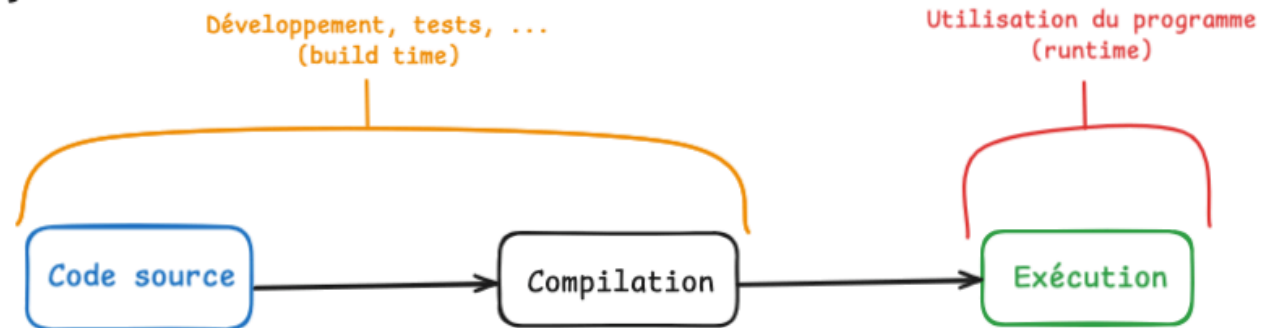


Langage interprété ou compilé ?

Langage interprété (R, Python, ...)



Langage compilé (Rust, C++, ...)



Memory safety

```
1 fn main() {
2     let s1 = String::from("hello");
3     let s2 = s1;
4
5     println!("{}", s1);
6     println!("{}", s2);
7 }
```

```
1 error[E0382]: borrow of moved value: `s1`
2 --> /Users/josephbarbier/l/talk/src/rencontresR2026/script.rs:5:20
3 |
4 2 |     let s1 = String::from("hello");
5   |         -- move occurs because `s1` has type `String`, which does not implement the `Copy` trait
6 3 |     let s2 = s1;
7   |         -- value moved here
8 4 |
9 5 |     println!("{}", s1);
10  |         ^^ value borrowed here after move
11 |
12 help: consider cloning the value if the performance cost is acceptable
13 |
14 3 |     let s2 = s1.clone();
15  |         +++++++
```

Type safety

```
1 fn main() {  
2     let x: i32 = 10;  
3     let y: &str = "20";  
4  
5     let result = x + y;  
6 }
```

```
1 error[E0277]: cannot add `&str` to `i32`  
2 --> /Users/josephbarbier/l/talk/src/rencontresR2026/script2.rs:5:20  
3 |  
4 5 |     let result = x + y;  
5 |                       ^ no implementation for `i32 + &str`  
6 |  
7 = help: the trait `Add<&str>` is not implemented for `i32`  
8 = help: the following other types implement trait `Add<Rhs>`:  
9     `&i32` implements `Add<i32>`  
10    `&i32` implements `Add`  
11    `i32` implements `Add<&i32>`  
12    `i32` implements `Add`
```

- “Oui mais en R aussi j’aurai une erreur”

Expérience développeur

Ecosystème Python

- `uv`: package/project manager
- `ruff`: linter & formatter
- `ty`: type checker
- `pyrefly`: type checker
- `polars`: data manipulation
- `pydantic`: data validation
- `monty`: interpreter
- `prek`: pre-commit hook
- **et plein d'autres!**

R ❤️ Rust : ça existe déjà
(et ça marche)

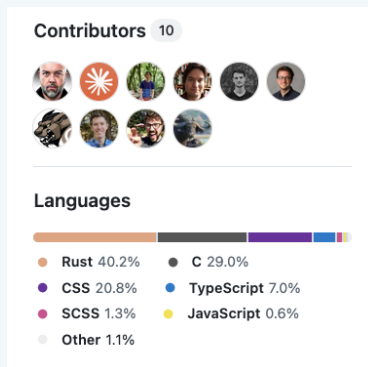
Rust dans l'écosystème R

- `Ark`: kernel R (utilisé dans Positron IDE)
- `Air`: formatter et serveur de langage
- `Jar`: linter
- `yaml12`: YAML parser
- `polars`: R interface à polars
- **et plein d'autres !**

Projet en développement

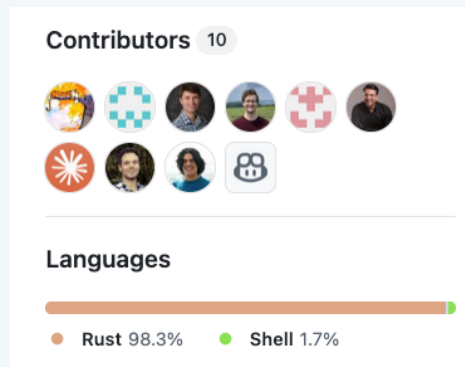
q2

Quarto version 2



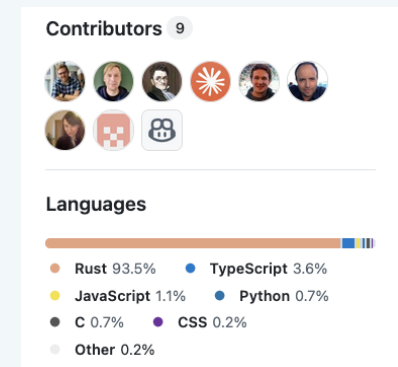
rv

Gestion des dépendances
et paquets R



ggsql

Grammaire des graphiques
et SQL



Appeler Rust depuis R : [extendr](#)

Build blazingly fast R packages with Rust

Use tools that put developers first.
Get support for publishing to CRAN.

Tests failing crates.io v0.9.0 docs passing License MIT

`$ init` `$ ls examples/` `$ --help`

Get Started

Build your first Rust-powered R package in minutes.

```
# setup R package
usethis::create_package("helloextendr")

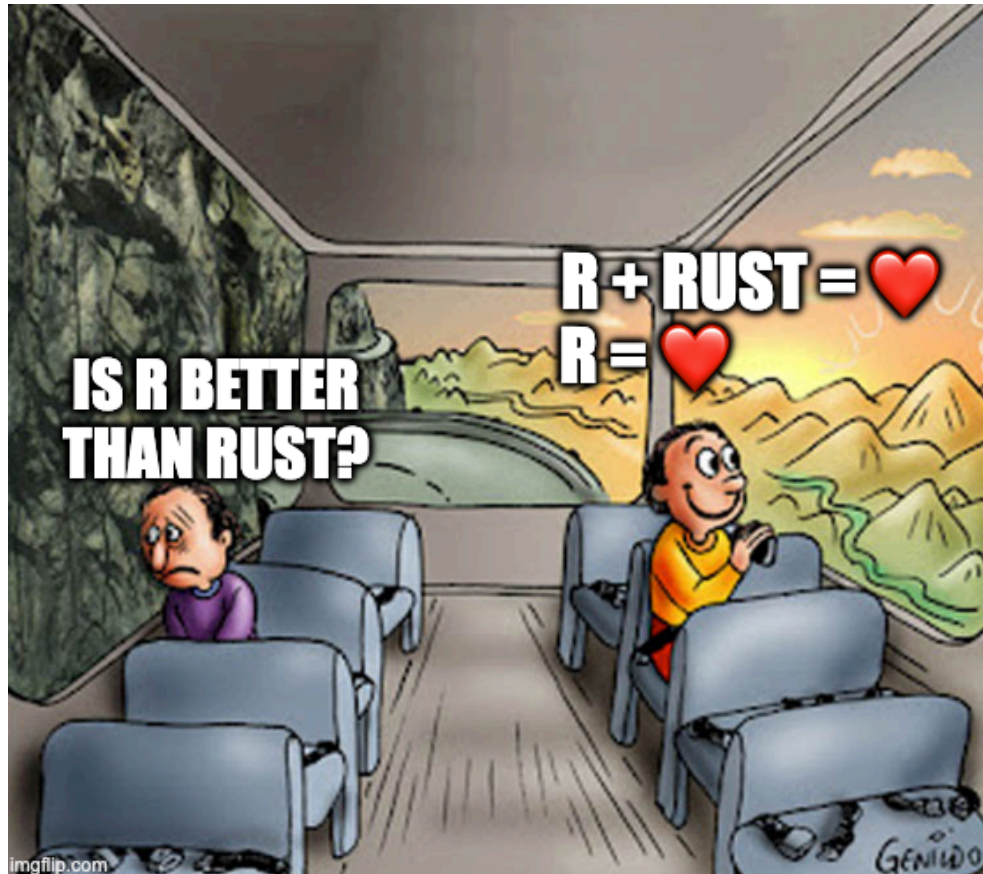
# add extendr scaffolding
rextendr::use_extendr()

# build and document package
devtools::document()
```

```
1 use extendr_api::prelude::*;
2
3 /// @export
4 #[extendr]
5 fn hello() -> &'static str {
6     "Hello RencontresR 2026 !"
7 }
8
9 extendr_module! {
10     mod helloextendr;
11     fn hello;
12 }
```

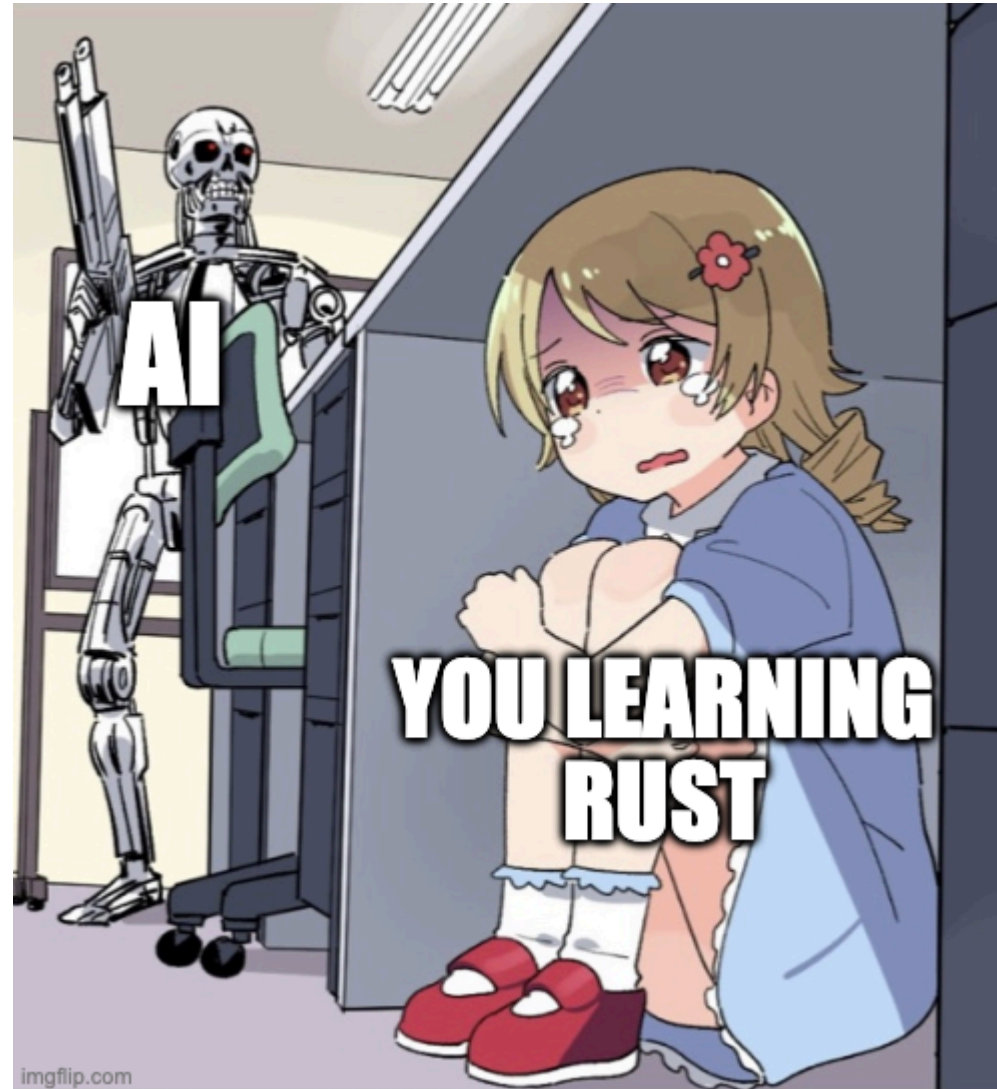
```
1 devtools::document() # build package
2 devtools::load_all() # load package
3
4 hello()
5 #> "Hello RencontresR 2026 !"
```

Est-ce que je dois arrêter R pour faire du Rust ?



- Rust ne remplace(ra) pas R
- Rust fonctionne **avec** R
- Rust peut remplacer C/C++
- Rust ne résoudra pas tous vos problèmes

Un mot sur l'IA



Merci !

J'ai dit des choses fausses ?

- venez m'en parler !

J'ai dit des choses pertinentes ?

- venez m'en parler !

Autres ?

- venez me voir !
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- slides: github.com/JosephBARBIERDARNAL/talk